HealthConnect Analytics — Part 2: Refinement & Business Insights

AnalystLab Africa — Data Analytics Track | Week 7 Deliverable

PROJECT	PHASE	TRACK	STATUS
HealthConnect	Part 2 Refinement	Data Analytics	Completed

1. Analytical Validation & KPI Results

- **Data Consistency Verification:** Cross-verified patient attendance and appointment volume across underlying database tables (N = 4,400 total records), confirming zero record loss during transformations.
- **DAX Metric Corrections:** Audited and corrected Total Department measure from 5 to a dynamic count of **200 departments/units**.
- **Data Model Relationship Audit:** Validated the 1-to-many relationship between Appointment and Service tables for complete cross-filtering.

2. Validated Key Findings

- **Laboratory Test Utilization:** Confirmed **Malaria testing** is the single highest volume laboratory procedure. Strategic inventory planning should prioritize malaria test kits.
- **Operational Footprint:** Validated operational footprint of 200 total departments/units across network branches.
- **Service & Appointment Cohesion:** Alignment between clinical services and appointments provides full visibility into patient throughput.

3. Dashboard Refinement & Usability Enhancements

- **Visual Precision:** Re-mapped Laboratory Test Distribution visual canvas to ensure visual fidelity and correct categorical ranking.
- **Slicer UX Cleansing:** Stripped null and blank values from Room Name slicers and restructured Service Name slicers for executive filtering.

4. Cross-Track Testing Contribution

- **Data Science Track:** Supplied clean, validated metrics to support the predictive no-show classification model training.
- **Generative AI Track:** Ensured calculated KPIs align with HealthConnect Knowledge Base reference standards.

5. Remaining Analytical Limitations

- **Static Retrospective Scope:** Current reporting relies on batch sample extracts; real-time tracking requires direct EHR API connections.
- **Unmapped External Factors:** External variables such as patient travel distance and local weather disruptions are unmapped.

6. Week 8 Integration Recommendations

- **Automated Data Validation:** Implement automated DAX assertion checks prior to final dashboard integration.
- **Live EHR Integration:** Transition data feeds to direct API/database replication pipelines before final deployment.